GENERATIVE AI & AGENTS

AI · ML · DS / REVISION NOTES

COURSE NOTES

GenAI, Agentic AI & Prompt Engineering

What generative AI is, agentic patterns and prompting ChatGPT well

5
SECTIONS

6
PAGES

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github.com/ankitgupta1729/AI-ML-DS

Contents

2 Core Technologies in Generative AI 3
3 Foundational Breakthroughs 3
4 The GPT Revolution 4
9 Role Prompting 5
10 Chain of Thought Prompting 5
LECTURE 11 6

1. Generative AI represents a revolutionary subfield of AI. It focuses on creating entirely new content including text, image and audio. The technology uses deep learning to mimic patterns from existing data.

Definition of Generative AI:

- Novel Data Creation:

AI branch dedicated to generating new data based on training inputs

- Pattern Mimicking:

Produces outputs similar to but distinct from original data

- Multi-Modal Output:

Creates text,images, code, music and other content types.

TOPIC 2

Core Technologies in Generative AI

A. Large Language Models (LLMs):

Advanced text generation like ChatGPT

B. Generative Adversarial Networks (GANs):

Competing networks create realistic outputs

C. Neural Network foundations:

Deep learning trained on massive datasets

TOPIC 3

Foundational Breakthroughs

A. Variational Autoencoder (VAE) (2013):

VAE technology introduced for data generation

B. Generative Adversarial Networks (GAN) (2014):

GANs creation, revolutionizing image generation

C. Diffusion Models (2015):

Incorporating noise into training data

D. Transformer Architecture (2017):

Proposed framework for modern AI systems

TOPIC 4

The GPT Revolution

A. GPT Launch (2018):

OpenAI introduces breakthrough large language model

B. Dall-E Debut (2021):

Revolutionary AI artwork generation capabilities

C. Public Release (2022):

Stable diffusion and midjourney democratize access

For multimodal AI, GPT-4o combines text, audio and visual processing.

5. ChatGPT is a state-of-the-art language processing AI developed by OpenAI.

ChatGPT uses a Generative pre-trained transformer architecture which is highly effective in understanding and generating human like text.

6. Langchain enables the creation of powerful agents that can perform the complex tasks autonomously. These agents can access external tools like databases, APIs and search engines together and take actions.

Langchain is an open-source framework which is build for LLM-powered applications and developer-friendly workflows. It provides the multi-language support for Python, JavaScript, typescript etc. Langchain was released in 2022 by Harrison Chase and Ankush Gola.

7. What are AI agents ?

A. Autonomous Intelligence:

Semi-autonomous tools powered by LLMs. Make independent decisions and execute complex tasks.

B. Workflow Automation:

Automate complex workflows and interact with external tools. Connect multiple APIs seamlessly.

C. Beyond Chatbots:

Go beyond simple conversations. Central component in modern GenAI applications.

8. What is Agentic AI ?

A. Autonomous Operations:

AI that acts independently with minimal human supervision required.

B. Goal-Oriented Behavior:

Achieves specific objectives without constant human guidance or intervention.

C. Independent Actions:

Moves beyond traditional rule-following to make sophisticated decisions autonomously.

TOPIC 9

Role Prompting

Assign specific personas for targeted outputs. Enhances task-specific performance significantly.

TOPIC 10

Chain of Thought Prompting

A. Problem Analysis:

Break complex tasks into logical steps

B. Sequential Reasoning:

Guide AI through explicit thought process

C. Improved Accuracy:

Significantly better results on complex problems

D. Real Examples:

Explaining neural network learning mechanism

- Key techniques:

A. Self-Consistency Method:

Generate multiple responses to same prompt. Select highest quality answer through comparison. Reduces randomness significantly.

B. Template Prompting:

Standardized formats for repeated tasks. Ensures consistency across similar requests. Saves time and improves reliability.

C. Combined benefits:

Bias reduction and increased output quality. Essential for production environments requiring consistency.

Lecture 11